AKSHAR VANDARA

Boston, MA | (857) 919-6234 | axr230102@gmail.com | LinkedIn | GitHub | Portfolio

Software/Research Engineer • 3 peer-reviewed publications in PCG and constraint optimization

EDUCATION

Northeastern University — *M.S., Game Science and Design* **May '26**
GPA: 3.87 / 4.0 | Coursework: Game Data Science, Game AI *Boston, MA*

Gujarat Technological University — *B.E., Computer Engineering* **May '23**
GPA: 3.6 / 4.0 | Coursework: Algorithms, AI, ML *Ahmedabad, India*

SKILLS

Languages:	C++, C#, Python, JavaScript/TypeScript, Java, SQL, R, C
Game Dev:	Unity (2D/3D), Unreal Engine 5, Blueprints, Behavior Trees, EQS, Perforce, ParaView/VTK
ML / AI:	PyTorch, TensorFlow, NumPy, Pandas, CUDA, Z3 SMT, PySAT, SAT/CSP modeling, OpenCV
Backend / Web:	REST APIs, Flask, Django, Node.js, React, Firebase, AWS, GCP, Git, Docker
CS Foundations:	Data Structures, Algorithms, Object-Oriented Design, Distributed Systems, Multithreading

EXPERIENCE

Research Assistant **Feb '25 – Present**
Game Design Lab, Northeastern University *Boston, MA*

- Engineered constraint-solving level-generation pipeline called Sturgeon (**PySAT**) that generates 2D puzzle levels with valid, solvable paths, cutting generation time **100–500×** vs. Spacetime WaveFunctionCollapse (*first-author, AAAI AIIDE 2025*).
- Designed an interactive editor for real-time level/path editing, plus **3D voxel visualization** via VTK/ParaView.
- Extended the system to **4D spacetime blending** of learned game mechanics (*first-author, FDG 2026 PCG Workshop*).
- Built a Python constraint solver (**Z3 SMT**) for generating **continuous spacetime** gameplay (*first-author, FDG 2026 PCG Workshop*).

Gameplay Systems & UI Programmer **May '25 – Aug '25**
Otisco Studios *Orlando, FL*

- Programmed **8+ core gameplay systems** in Unreal/C++ (including save/load and UI) across two titles.
- Created **5+ NPC** and enemy AI archetypes for roguelike and open-world genres using C++, Behavior Trees, and **custom EQS**.
- Resolved **30+** critical pre-launch bugs to stabilize the build for release.
- Profiled and refactored gameplay code, **doubling average framerates (2x)**.

Unity Developer **Mar '23 – Jul '24**
Arcadon Games *Bangalore, India*

- Developed core gameplay mechanics in Unity/C# using state machines, improving input **responsiveness 30%.**
- Integrated Firebase to store and sync **1,000+** user profiles and inventory data, with sub-100ms response.
- Wrote a match-simulation system that processes **30 matches/min** and cuts load times 40%.
- Ran alpha testing with **100+ users (85% satisfaction)**; fixed 20+ stability bugs before launch.

PROJECTS

Spacetime PCG Research Suite [GitHub | Publication] | ***Python, Z3 SMT, PySAT, Pandas, Paraview***

- Open-sourced an extension of Sturgeon for spacetime-constrained level generation; supports 2D, 3D-blended, and continuous physics-based platformer domains.
- Benchmarked generation speed and solvability across the three domains with Pandas, surfacing per-domain performance trade-offs.

Red Circuit Rebellion [Steam] | ***Unreal Engine 5, C++, Blueprints, Behavior Trees, EQS***

- Shipped a title on Steam with two distinct game modes (Story and Arcade), each with its own objective logic and enemy spawners.
- Implemented standout gameplay features in C++ and Blueprints, including real-time environmental fracturing and a dynamic objectives manager.

Cricket Manager: League [Playstore] | ***Unity, C#, Firebase, Cloud Functions***

- Architected sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features.
- Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.